2	**(a)**	**(i)**	number of <u>atoms/nuclei</u> in 12 g of carbon-12	B1	[1]
		(ii)	amount of substance	M1	
			containing N_A (or 6.02×10^{23}) particles/molecules/atoms *or* which contains the same number of particles/atoms/molecules as there are atoms in 12 g of carbon-12	A1	[2]
	(b)		$pV = nRT$		
			$2.0 \times 10^7 \times 1.8 \times 10^4 \times 10^{-6} = n \times 8.31 \times 290$, so $n = 149$ mol or 150 mol	A1	[1]
	(c)	**(i)**	V and T constant and so pressure reduced by 5.0% pressure $= 0.95 \times 2.0 \times 10^7$	C1	
			or		
			calculation of new n (= 142.5 mol) and correct substitution into $pV = nRT$	(C1)	
			pressure $= 1.9 \times 10^7$ Pa	A1	[2]

(ii)	loss is $5/100 \times 150\,\text{mol} = 7.5\,\text{mol}$ *or* $\Delta N = 4.52 \times 10^{24}$	C1	
	$t = (7.5 \times 6.02 \times 10^{23})/1.5 \times 10^{19}$ *or* $t = 4.52 \times 10^{24}/1.5 \times 10^{19}$	C1	
	$= 3.0 \times 10^{5}\,\text{s}$	A1	[3]